

Learning Rewrite-Invariant Reasoning with Targeted Alternation Training

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1. Motivation

Meaning-preserving rewrites should not change the answer. In practice, an LLM can solve one wording and fail on another wording of the same task.

Failure mode. A rewrite can push the sampled reasoning trajectory across a point after which the model no longer recovers.

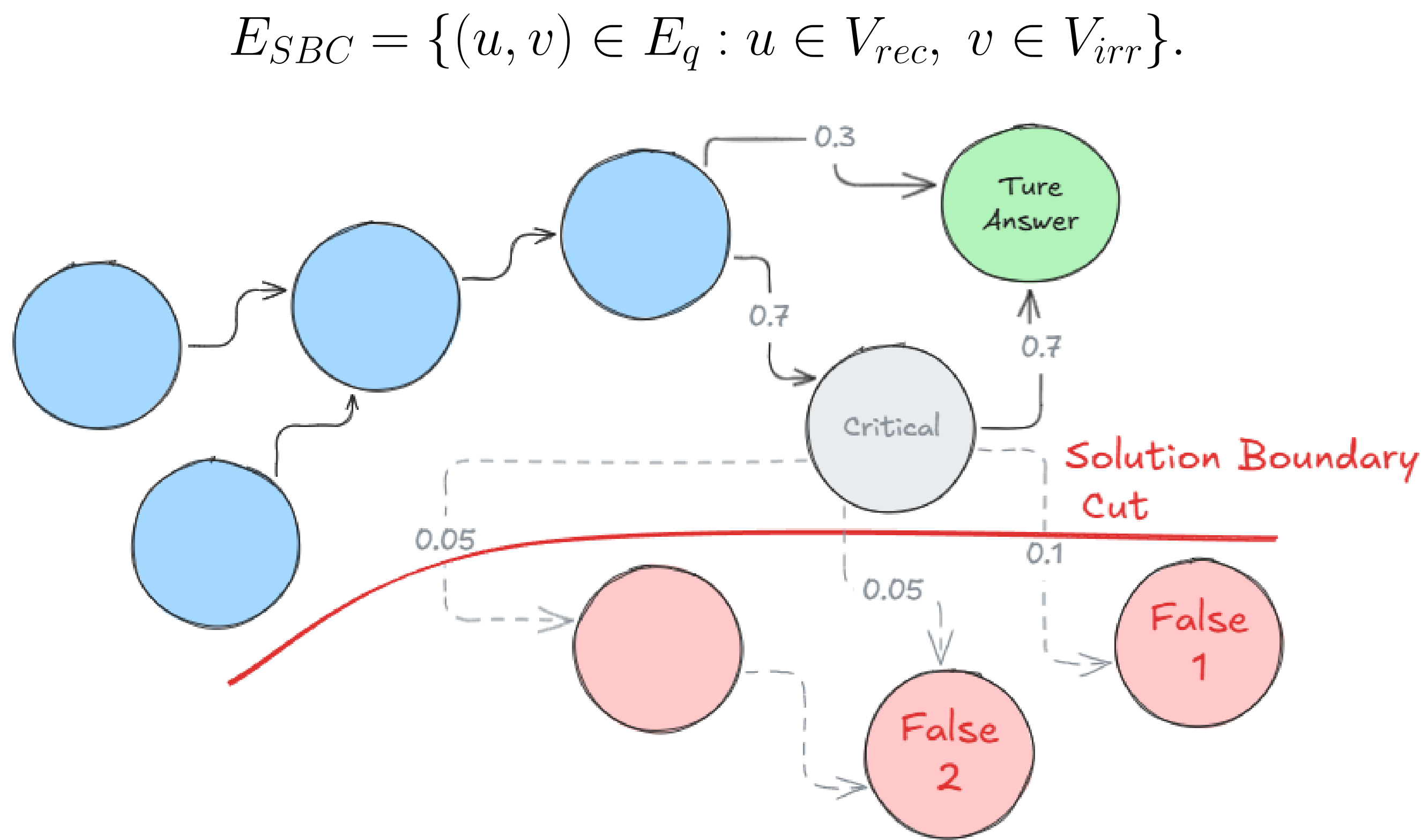
Training issue. Uniform paraphrase augmentation spends budget on easy or redundant rewrites rather than on the variants that repeatedly trigger recoverable-to-irrecoverable transitions.

2. Contributions

- Reasoning graphs:** aggregate traces from the original question and accepted rewrites into a per-question graph.
- Solution Boundary Cut:** identify edges that leave the recoverable region.
- Boundary predictor:** learn recurring, model-specific crossing patterns.
- Targeted alternation training:** convert high-confidence crossings into contrastive rewrite examples.

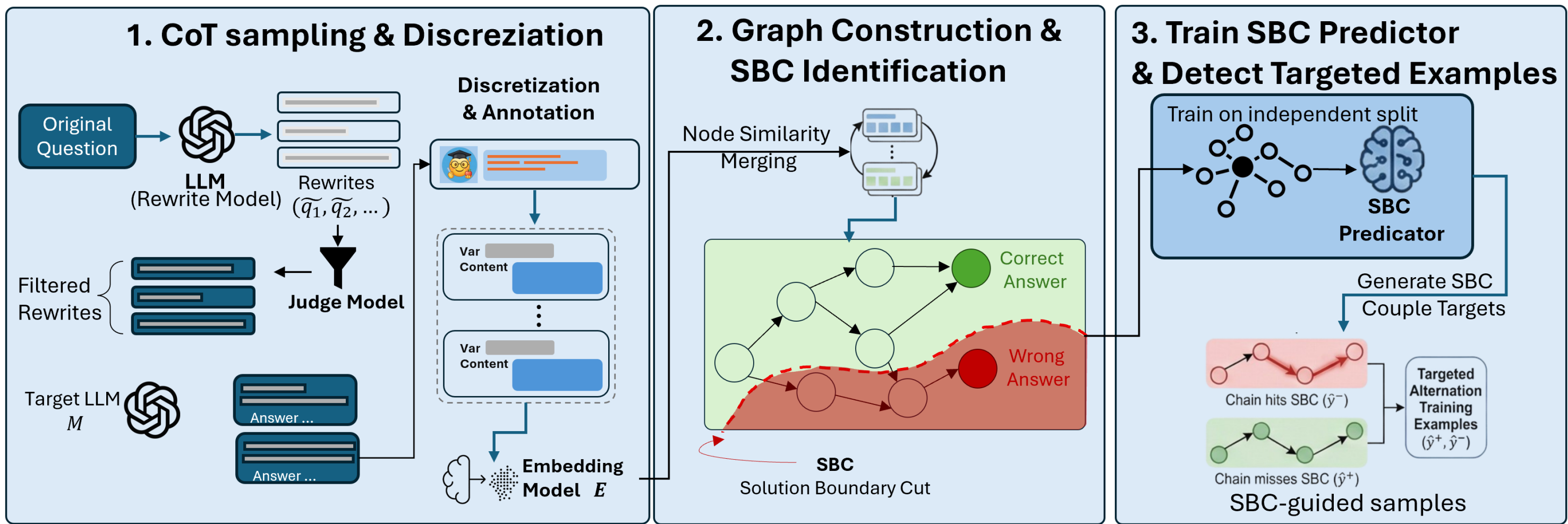
Core object: Solution Boundary Cut

A node is **recoverable** if it can reach a correct terminal answer; otherwise it is **irrecoverable**. The boundary contains edges from recoverable to irrecoverable states:



SBC localizes the first observed transition after which sampled reasoning does not recover.

3. Paper idea



Pipeline: sample rewrites, build reasoning graphs, detect SBC edges, and train on targeted pairs.

- Sample.** Accepted rewrites and target-model traces.
- Discretize.** Comparable steps with operation and variable tags.
- Build graph.** Merged states and observed transitions.
- Locate SBC.** Reverse reachability from correct terminals.
- Train.** DPO pairs, or in-context examples for GPT-5.2.

Protocol. Source benchmarks are MMLU-Pro, Big-MATH, and DROP; HLE is transfer-only. Each base question uses five accepted rewrites and $K = 2$ traces for the original plus each rewrite, yielding **12 traces**. Evaluation is leave-one-benchmark-out.

5. Ablations

Alternations matter

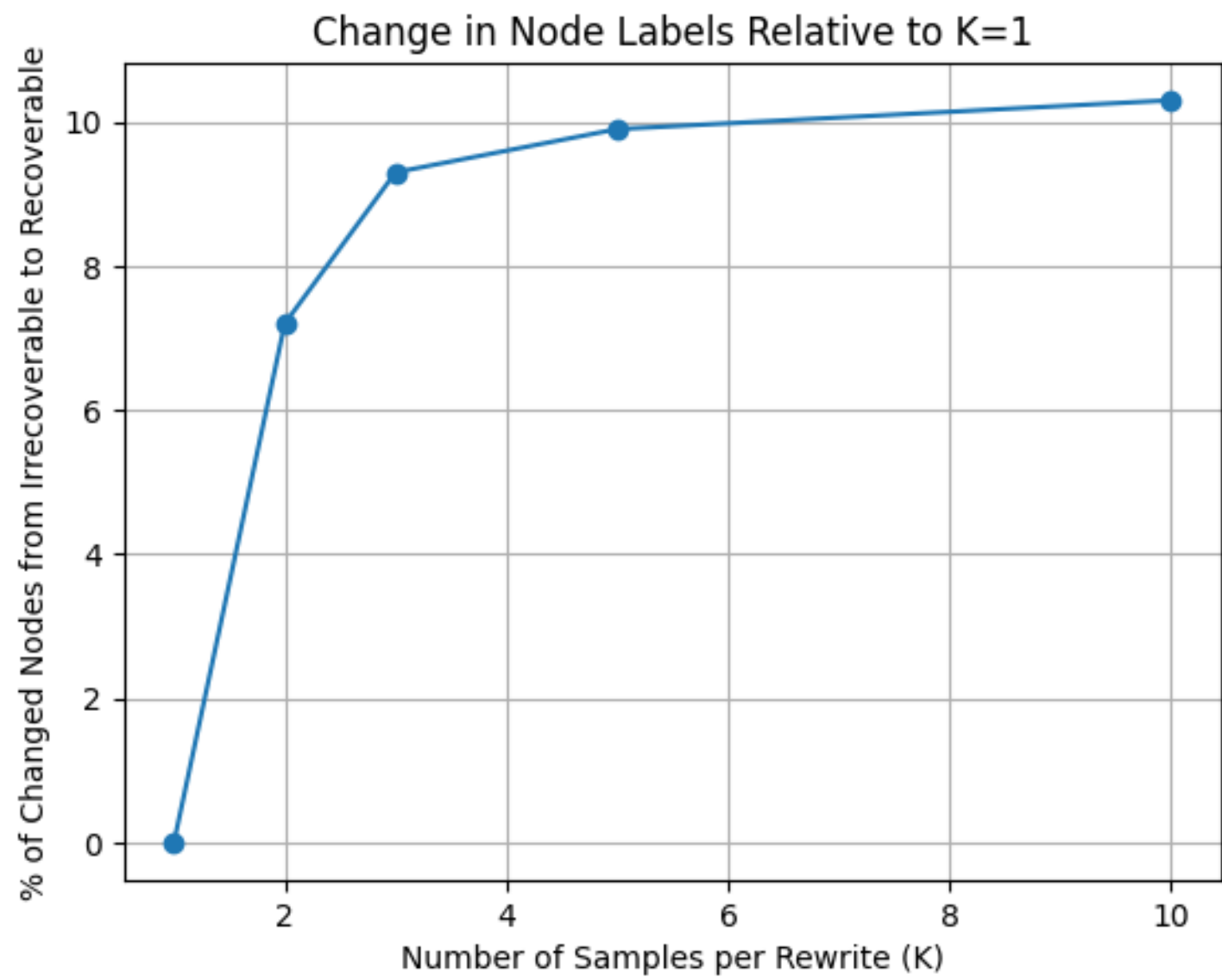
Setting	B-M	M-P	DROP
Base	67.1	76.5	85.9
No alt.	66.7	76.6	86.1
With alt.	74.8	78.0	91.9

Node annotations improve merging

	No tag	Tag
No op.	0.87	0.91
Op.	0.92	0.94

Graph structure helps

	Pred.	DS	GPT	LL	Phi
LogReg	70.4	70.6	69.9	72.3	
G-Trans	75.4	74.6	73.7	77.2	
GNN	80.2	78.1	76.9	83.8	



$K = 2$ captures most recoverability-label corrections.

4. Results

+4.5–5.1 average points over base across source benchmarks
35.4 → 38.1 GPT-5.2 xhigh on HLE with 200 examples
12 sampled traces per base question

Main results table. Targeted alternation training is best or tied-best on nearly all model/benchmark combinations under leave-one-benchmark-out training.

Model	Train	Big-Math	MMLU-PRO	DROP
DeepSeek-V2-Lite	Base	52.7 ± 0.61	60.1 ± 1.04	48.1 ± 1.15
	SFT	54.3 ± 0.61	53.8 ± 1.19	49.8 ± 1.28
	Hard	54.4 ± 0.61	60.3 ± 1.34	50.6 ± 1.43
	Random	54.6 ± 0.62	54.5 ± 1.27	50.0 ± 1.35
	TT vote	53.7 ± 0.61	60.6 ± 1.43	51.2 ± 1.51
	Positive	56.0 ± 0.60	59.6 ± 1.26	53.1 ± 1.35
	Targeted	58.1 ± 0.59	61.8 ± 1.26	54.4 ± 1.35
Phi-4	Base	63.4 ± 0.58	71.1 ± 0.96	76.8 ± 0.98
	SFT	65.0 ± 0.58	70.2 ± 1.11	76.4 ± 1.10
	Hard	65.1 ± 0.59	70.5 ± 1.26	75.7 ± 1.23
	Random	65.0 ± 0.51	70.8 ± 1.19	75.0 ± 1.19
	TT vote	64.2 ± 0.59	71.5 ± 1.34	82.2 ± 1.26
	Positive	67.2 ± 0.57	71.8 ± 1.19	81.4 ± 1.23
	Targeted	68.2 ± 0.56	72.3 ± 1.18	84.2 ± 1.15
LLaMA-3.3-70B	Base	61.5 ± 0.58	71.9 ± 0.96	67.1 ± 1.07
	SFT	62.0 ± 0.57	72.1 ± 1.11	67.9 ± 1.13
	Hard	62.2 ± 0.59	69.7 ± 1.28	71.5 ± 1.20
	Random	62.3 ± 0.59	69.6 ± 1.28	69.0 ± 1.16
	TT vote	61.9 ± 0.52	73.0 ± 1.33	74.1 ± 1.26
	Positive	64.4 ± 0.53	71.8 ± 1.19	74.2 ± 1.28
	Targeted	65.3 ± 0.58	72.8 ± 1.18	76.5 ± 1.24
GPT-4.1-mini	Base	67.1 ± 0.51	76.5 ± 0.91	85.9 ± 0.87
	SFT	69.9 ± 0.56	76.7 ± 1.05	86.4 ± 0.96
	Hard*	70.0 ± 0.56	76.7 ± 1.27	86.5 ± 1.23
	Random	70.5 ± 0.56	77.2 ± 1.18	86.0 ± 1.14
	TT vote	69.1 ± 0.57	77.0 ± 1.36	85.0 ± 1.34
	Positive	70.5 ± 0.56	77.8 ± 1.17	90.1 ± 1.09
	Targeted	74.8 ± 0.53	78.0 ± 0.86	91.9 ± 0.89

Accuracy (%) with estimated 95% confidence intervals. SFT = standard task supervision; Hard = hard-example mining; TT vote = inference-time rewrite ensembling.